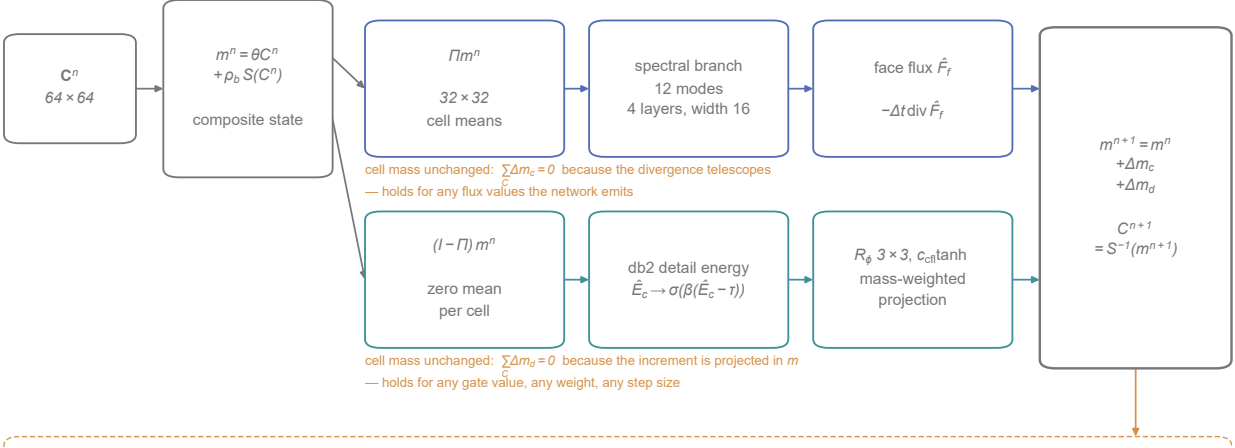

C^n
64×64
$m^n = \theta C^n$
$+ \rho_b S(C^n)$
composite state
Πm^n
32×32
cell means
spectral branch
12 modes
4 layers, width 16
face flux $\hat{F}_f$
$-\Delta t \operatorname{div} \hat{F}_f$
$m^{n+1} = m^n$
$+\Delta m_c$
$+\Delta m_d$
C^{n+1}
$= S^{-1}(m^{n+1})$
cell mass unchanged: $\sum_C \Delta m_c = 0$ because the divergence telescopes
— holds for any flux values the network emits
$(I - \Pi) m^n$
zero mean
per cell
db2 detail energy
$\hat{E}_c \to \sigma(\beta(\hat{E}_c - \tau))$
R_ϕ 3×3, c_{cft} tanh
mass-weighted
projection
cell mass unchanged: $\sum_C \Delta m_d = 0$ because the increment is projected in m
— holds for any gate value, any weight, any step size
audited budget on $M^n = \sum_i m_i^n \Delta A$: $r(t) = \left| M^n - M^0 - \sum_k \Delta t (\hat{S}^k - \lambda M^k) \right| / M^0$
r is identical across the conserving arms (6.1×10^{-6}, the audit's own floor) and $O(1)$ without the flux form
— the source cancels term by term, so $r(t)$ can respond to the operator and never to the gate